



EE/CE 112: Electric Circuits-I

Syed Arsalan Jawed & Saad Baig

Homework 2

Assigned: September 20th, 2024

Due by: September 28th, 2024

Total marks: 40

Important Notes

There are problems for which pen and paper solutions have to be submitted for grading. Preceding the solutions, you must include the following information:

- Your name
- Student ID
- Course ID
- Homework number
- Date and time submitted

Solution should be elaborative, sketch neat circuits wherever required, show clearly all steps of calculation & mention reference (for example formula, law, rule) to strengthen your understanding to solve the electric circuit problems.

Also, please comply with Academic Integrity Policies of Habib University related to plagiarism while submitting this assignment for grading.

Late Submission Policy

You need to submit your solution by the due date. Every day the assignment is late, 10% will be reduced from the total score. Assignment will not be graded after fifth day of submission deadline.

Course Learning Outcomes targeted using this assessment		
CLO	Description	Learning-Domain Level
CLO 2	Apply nodal and mesh analysis techniques in the identification, formulation, and solution of the basic problem of circuit analysis.	Cog - 3

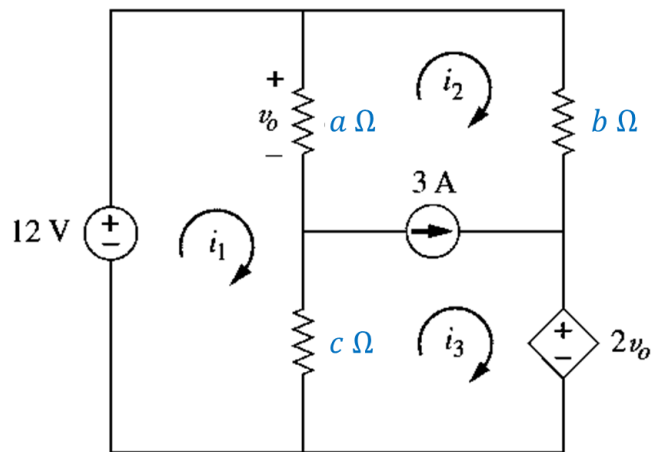


NOTE: Some questions are randomized through your HU ID. Let v , w , x , y and z be the 5 digits of your ID. Then consider the following mapping to attempt questions:

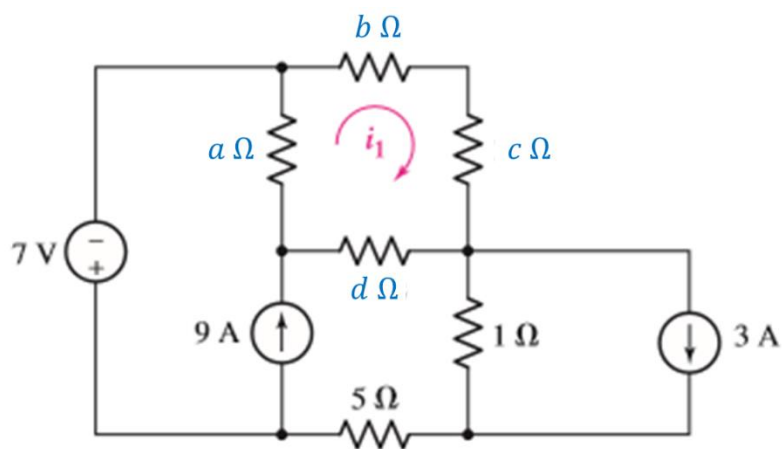
1. $a = v + w$
2. $b = v + w + x$
3. $c = v + w + x + y$
4. $d = v + w + x + y + z$

As an example, for HU ID 01234, $a = 1$, $b = 3$, $c = 6$ and $d = 10$.

Question 1 [10 pts]: Calculate all mesh currents for the circuit below, creating supermeshes wherever needed.

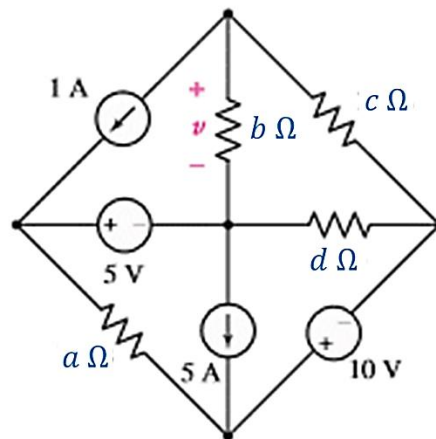


Question 2 [10 pts]: Calculate all mesh currents for the circuit below, creating supermeshes wherever needed.





Question 3 [10 pts]: Calculate all mesh currents for the circuit below, creating supermeshes wherever needed.



Question 4 [10 pts]: For the circuit below, write all mesh equations in terms of the mesh currents. No need to solve the system of equations.

